Examiner reports

Q1.

This question was answered well.

Q2.

Most candidates answered this question well, with only a few making algebraic or arithmetical errors.

Q3.

While candidates realised that part (a) required differentiation, a significant number were not able to carry this out accurately. Part (b) proved more successful, but only a small minority of candidates were able to attempt part (c), the simplest solution involving maximisation of the appropriate trigonometrical function.

Q4.

Quite a few candidates had problems dealing with the differentiation and integration of the exponential function in this question. They used techniques that were confused and were unable to obtain the correct results. This was probably due to the fact that this question demanded the use of techniques from the core mathematics modules, with which the candidates were not yet fully confident. In part (a)(ii), very few candidates gave fully correct answers for the range of values of the acceleration. Some obtained either the value 2 or the value 14. Of those that did obtain both values, very few used the correct inequalities. In part (b), there were problems with the integration of the exponential function and also with candidates omitting or not find the constant of integration.

Q5.

Part (a) was frequently done well, with the most common error in part (a)(ii) being the introduction into the equation of motion of an additional force, usually the weight of the motorcycle and rider. Part (b) seldom produced assumptions backed by sound reasons. The majority of responses referred to information given in the question. In part (c), the most convincing solutions were based on the algebraic link between the force and the radius. There was a tendency in part (c) to assume that the magnitude of the frictional force could be altered by changing the radius of the circle, rather than the necessity to adjust the radius to compensate for the force.

Q6.

Many candidates proved competent in handling vector expressions, but a significant number lacked appreciation of vector and scalar quantities and the necessary notation. Errors in differentiation of trigonometrical functions, particularly in signs, also spoiled otherwise excellent solutions. Otherwise part (a) was done well, with the most frequent error being in the failure to present an answer in radians in part (a)(iii). Responses in part (b)(i) were variable, with scalar answers frequently presented, while in part (b)(ii), few appreciated the variable nature of the direction of the force, and many seeming unaware of both upper and lower bounds of the magnitude of the force.

Q7.

Intended as an easy starter, responses to this question were surprisingly poor. Many candidates failed to appreciate the need to differentiate the given expression for velocity, assuming the constant acceleration equations to provide an appropriate method of solution.

Q8.

Parts (a), (b) and (c) were generally answered very well.

In part (a), a few candidates tried to substitute $t = 0$ into the position vector without differentiating.

In part (c), some candidates found the acceleration, but did not go on to find the magnitude.

Part (d) was more challenging. Quite a few candidates stated that the velocity would become zero. There were some very good answers, but also quite a few that were somewhat confused.

Q9.

The candidates generally did well on this question, although some did make errors when differentiating or substituting for t. Part (b) was found to be more difficult and some candidates produced very confused solutions.

Q10.

Part (a) was answered correctly by almost all of the candidates. While some went on to deal correctly with part (b), a large number of candidates worked with the wrong component. Part (c) was found to be more challenging and there were very few correct solutions. Most of the candidates worked with the position vector instead of the velocity vector.

Q11.

This question was generally done very well. The only significant errors were associated with differentiation techniques. A few candidates did not multiply by –1 when differentiating $2e^{-t}$. A few also used some unconventional rules that seemed to be based on the product rule.

Q12.

Part (a) of this question was done well by many candidates. The main reason that candidates lost marks was because they did not differentiate correctly. For parts (b) and (c), many candidates used the position vector rather than the velocity vector as the basis for their working. This fundamental error was very common. In part (c), a number of candidates worked in degrees rather than radians, obtaining a final answer of 3600 instead of 20 π.